EDUCATIONAL MEASUREMENT & PSYCHOMETRICS

Estimating Test Reliability: Methods & Sample Computations

A practical guide to Test-Retest, Parallel-Forms, Split-Half, and Internal Consistency formulas with step-by-step mathematical solutions.

OVERVIEW

Four Core Reliability Frameworks

Test-Retest

Temporal stability across administrations.

Parallel-Forms

Equivalence across alternate instruments.

Split-Half

Spearman-Brown internal test partitioning.

KR-20 Method

Homogeneity across dichotomous items.

METHOD 01

Test-Retest Method & Spearman Rank Formula

The identical assessment instrument is administered twice to the same cohort over a predetermined time interval.
When participant scores are expressed as ordinal rankings, Spearman's rank correlation coefficient calculates the consistency of ranking stability.

SPEARMAN RANK CORRELATION FORMULA

$$r_s = 1 - \frac{6 \sum D^2}{N^3 - N}$$

Where **D** represents rank differences and **N** is sample size.

WORKED EXAMPLE

Test-Retest: Step-by-Step Computation

Student	Rank 1	Rank 2	D (Diff)	D^2
A	1	1	0	0
B	2	3	-1	1
C	3	2	+1	1
D	4	4	0	0
E	5	5	0	0
Total ΣD^2	**2**			

Evaluation (N = 5)

Denominator: $5^3 - 5 = 125 - 5 = 120$

Numerator: $6 \times \Sigma D^2 = 6 \times 2 = 12$

Fraction: $12 \div 120 = 0.10$

$r_s = 1 - 0.10 = 0.90$

High temporal rank stability.

METHOD 02

Parallel-Forms: Equivalent Test Administration

Two equivalent test forms with identical blueprints are administered to the same group to gauge form equivalence. Paired observation scores are correlated using the Pearson product-moment correlation formula.

PEARSON PRODUCT-MOMENT CORRELATION

$$r = \frac{N\sum XY - (\sum X)(\sum Y)}{\sqrt{[N\sum X^2 - (\sum X)^2][N\sum Y^2 - (\sum Y)^2]}}$$

Where **X** is Form A and **Y** is Form B.

WORKED EXAMPLE

Parallel-Forms: Step-by-Step Computation

Student	Form A (X)	Form B (Y)	X^2	Y^2	XY
1	10	9	100	81	90
2	8	8	64	64	64
3	6	7	36	49	42
4	5	4	25	16	20
5	3	4	9	16	12
Total	**32**	**32**	**234**	**226**	**228**

Formula Substitution

Numerator: $5(228) - (32)(32) = 1140 - 1024 =$ **116**

Denominator: $\sqrt{[(1170 - 1024)(1130 - 1024)]}$

Denominator: $\sqrt{(146 \times 106)} = \sqrt{15476} \approx$ **124.40**

$r = 116 \div 124.40 =$ 0.93

High parallel-form equivalence.

METHOD 03

Split-Half Method & Spearman-Brown Correction

A single test is split into equal halves (commonly odd vs. even numbered questions) and correlated.

Because dividing test length underestimates full test reliability, the Spearman-Brown prophecy formula restores full-scale estimation.

SPEARMAN-BROWN PROPHECY FORMULA

$$r_{wt} = \frac{2\,(\,r_{ht}\,)}{1 + r_{ht}}$$

Where **rwt** is whole-test reliability and **rht** is half-test correlation.

WORKED EXAMPLE

Split-Half: Step-by-Step Computation

Step 1

Odd-Even Partition

Correlated half-test correlation rht = 0.70.

Step 2

Formula Substitution

rwt = 2(0.70) / (1 + 0.70).

Step 3

Fraction Resolution

1.40 ÷ 1.70 = 0.824.

Calculated Whole-Test Reliability

Spearman-Brown correction increases reliability from 0.70 to 0.82.

rwt = 0.82

METHOD 04

Internal Consistency: Kuder-Richardson Formula 20

KR-20 evaluates item homogeneity across tests containing dichotomously scored items (scored right/wrong or 1/0). It calculates test variance against the sum of individual item variances across all test questions.

KUDER-RICHARDSON FORMULA 20 (KR-20)

$$R_{xx} = \frac{N}{N-1}\frac{SD^2 - \sum p_i q_i}{SD^2}$$

Where **N** = test items, **p** = pass rate, and **q** = fail rate (1 – p).

WORKED EXAMPLE

KR-20: Step-by-Step Computation

Item Count (N)	Total Variance (SD²)	Item Variances (Σ pq)
10	4.00	1.60

1. Scale Multiplier: [N / (N – 1)] = 10 ÷ 9 = **1.111**
2. Variance Ratio: [(4.00 – 1.60) / 4.00] = 2.40 ÷ 4.00 = **0.600**
3. Product: 1.111 × 0.600 = **0.667**

KR-20 RELIABILITY

0.67

EVALUATION GUIDE

Reliability Coefficient Benchmarks

Coefficient Range	Quality Rating	Appropriate Application
$r \geq 0.90$	Excellent	High-stakes diagnostic decisions.
$0.80 \leq r < 0.90$	Good	Classroom grading and certification.
$0.70 \leq r < 0.80$	Acceptable	Formative tests and research surveys.
$r < 0.70$	Questionable	Requires item revision or test lengthening.

SUMMARY & DISCUSSION

Measurement Precision in Practice

Reliability estimation provides the quantitative foundation for fair, defensible assessment design.